

Henry Harvin
YUVA INTERN PROGRAM

SQL DATA ANALYST - AGRIBUSINESS

Week 2 Project Report
Data Modeling & SQL Query Development for Agribusiness

Dataset:

Agribusiness Database - MySQL

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Week: 2

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Introduction

This project focuses on designing a relational database model for agribusiness operations and developing SQL queries to support common business and analytical needs.

The database is designed to represent different aspects of an agribusiness operation, including farm management, crop production, agricultural inputs, production costs, sales, markets, and buyers. The model is intended to support analysis of crop performance, farm operations, costs, revenue, profitability, and market trends.

The project is being developed using MySQL. The database design includes multiple related tables connected through primary and foreign keys, allowing the data to be stored in a structured and relational form.

Requirements Gathering

Before designing the database, the main business areas that the model should support were identified.

The database should support:

- **Farm management:** tracking farms and their locations.
- **Crop production:** recording crops, cultivated area, production quantity, yield, seasons, and production years.
- **Agricultural inputs:** recording inputs such as seeds, fertilizers, and pesticides and their usage.
- **Production costs:** tracking expenses related to production.
- **Sales:** recording crop sales, quantities, prices, and revenue.
- **Markets and buyers:** identifying where produce is sold and who purchases it.
- **Time and seasons:** supporting analysis across different years and agricultural seasons.
- **Business performance:** supporting analysis of production, yield, costs, revenue, and profitability.

The model should allow business questions such as:

- Which crops have the highest production or yield?
- Which farms have the highest production?
- How much does it cost to produce different crops?
- Which crops generate the most revenue or profit?
- Which markets generate the highest sales?
- How do production and revenue change over time?
- How do input costs relate to crop profitability?

Entity Identification

Based on the identified business requirements, the following entities were selected for the database:

Entity	Purpose
location	Stores state and district information
farm	Stores information about managed farms
crop	Stores crop information
season	Stores agricultural seasons
production	Records crop production activities
input	Stores agricultural input information
input_usage	Records inputs used for a production activity
cost	Records other production-related expenses
market	Stores information about selling markets
buyer	Stores buyer information
sale	Records crop sales and revenue

These entities were selected to represent the major operational areas of an agribusiness while keeping related information in separate tables.

Relationships Between Entities

The relationships between the entities were designed according to how the business operates.

A location can contain multiple farms, so there is a **one-to-many relationship** between location and farm.

A farm can produce multiple crops, and a crop can be produced on multiple farms. This many-to-many relationship is represented through the production table. Each production record connects a farm, crop, and season.

The production table also connects to input_usage and cost, allowing inputs and expenses to be associated with individual production activities.

Sales are connected to production, markets, and buyers. This allows the database to track where produce was sold, who purchased it, and the revenue generated.

The main relationships are:

location 1 ———< farm
farm 1 ———< production >———— 1 crop
season 1 ———< production

production 1 ———< input_usage >———— 1 input
production 1 ———< cost

location 1 ———< market
production 1 ———< sale >———— 1 market
buyer 1 ———< sale

Database Creation

A separate MySQL database was created for the Week 2 project so that it could be developed independently from the Week 1 database.

```
create database agribusiness_week2;  
use agribusiness_week2;
```

The database was designed with 11 relational tables. Each table has a primary key to uniquely identify its records, while foreign keys are used to connect related tables.

The tables were created in an order that allowed the required foreign-key relationships to be established correctly.

1) Location Table

The location table stores the state and district associated with a farm or market.

```
create table location (  
    location_id int primary key,  
    state varchar(100) not null,  
    district varchar(100) not null  
);
```

The location_id acts as the primary key and uniquely identifies each location.

2) Farm Table

The farm table stores information about farms and connects each farm to a location.

```
create table farm (  
    farm_id int primary key,  
    farm_name varchar(100) not null,  
    location_id int not null,  
    farm_area decimal(10,2) not null,  
    foreign key (location_id) references location(location_id)  
);
```

The location_id is a foreign key referencing the location table.

3) Crop Table

The crop table stores information about the crops used in the database.

```
create table crop (  
    crop_id int primary key,  
    crop_name varchar(100) not null,  
    crop_area decimal(10,2) not null,  
    foreign key (location_id) references location(location_id)  
);
```

```
crop_id int primary key,  
crop_name varchar(100) not null,  
crop_type varchar(100)  
);
```

4) Season Table

The season table stores agricultural season information.

```
create table season (  
    season_id int primary key,  
    season_name varchar(50) not null  
);
```

5) Production Table

The production table is the main table for recording crop production activities. It connects farms, crops, and seasons.

```
create table production (  
    production_id int primary key,  
    farm_id int not null,  
    crop_id int not null,  
    season_id int not null,  
    production_year int not null,  
    cultivated_area decimal(10,2) not null,  
    production_quantity decimal(12,2) not null,  
    yield_quantity decimal(12,2),  
    foreign key (farm_id) references farm(farm_id),  
    foreign key (crop_id) references crop(crop_id),  
    foreign key (season_id) references season(season_id)  
);
```

This structure allows production to be analysed by farm, crop, season, and year.

6) Input and Input Usage Tables

The input table stores agricultural inputs, while input_usage records how those inputs are used for a specific production activity.

```
create table input (  
    input_id int primary key,  
    input_name varchar(100) not null,  
    input_type varchar(50),
```

```

        unit varchar(30)
    );

create table input_usage (
    usage_id int primary key,
    production_id int not null,
    input_id int not null,
    quantity_used decimal(10,2) not null,
    cost decimal(12,2) not null,
    foreign key (production_id) references production(production_id),
    foreign key (input_id) references input(input_id)
);

```

This separation allows the same input to be used across multiple production activities.

7) Cost Table

The cost table stores other expenses associated with production.

```

create table cost (
    cost_id int primary key,
    production_id int not null,
    cost_type varchar(50) not null,
    amount decimal(12,2) not null,
    foreign key (production_id) references production(production_id)
);

```

8) Market and Buyer Tables

The market table stores selling markets, while the buyer table stores information about buyers.

```

create table market (
    market_id int primary key,
    market_name varchar(100) not null,
    location_id int not null,
    foreign key (location_id) references location(location_id)
);

```

```

create table buyer (
    buyer_id int primary key,
    buyer_name varchar(100) not null,
    buyer_type varchar(50)
);

```

The market is linked to the location table so that sales can also be analysed geographically.

9) Sale Table

The sale table records crop sales and connects each sale to its production activity, market, and buyer.

```
create table sale (  
    sale_id int primary key,  
    production_id int not null,  
    market_id int not null,  
    buyer_id int not null,  
    sale_date date not null,  
    quantity_sold decimal(12,2) not null,  
    selling_price decimal(10,2) not null,  
    revenue decimal(14,2) not null,  
    foreign key (production_id) references production(production_id),  
    foreign key (market_id) references market(market_id),  
    foreign key (buyer_id) references buyer(buyer_id)  
);
```

This structure allows sales and revenue to be analysed by crop, farm, market, buyer, and production activity.

Data Population

After creating the database structure and establishing the relationships between the tables, the next step was to populate the tables with realistic sample data. The data was designed to represent common agribusiness activities such as farm production, agricultural input usage, production costs, market transactions, and sales.

The sample data was kept manageable in size while providing enough variation for developing and testing practical SQL queries. Foreign key relationships were also maintained while inserting the data so that each record remained connected to the appropriate farm, crop, season, market, buyer, or production record.

1) Populated Tables

The following tables were populated:

Table	Number of Records
location	8
farm	8
crop	10
season	6
production	20
input	8
input_usage	48
cost	46
market	8
buyer	8
sale	28

The data covers multiple locations, farms, crops, agricultural seasons, production years, input types, cost categories, markets, and buyer types. This provides sufficient data for testing different types of business queries.

2) Data Population Approach

The tables were populated in an order that followed their dependencies. Reference tables such as location, crop, season, and input were populated first. Tables that depend on these reference tables, such as farm, production, input_usage, market, and sale, were then populated.

For example, the production table connects a farm with a crop, season, and production year. Similarly, the input_usage table connects agricultural inputs with individual production records.

The sale table contains transaction-level information including the market, buyer, quantity sold, selling price, and revenue. Revenue values were calculated using the relationship:

$$\text{Revenue} = \text{Quantity Sold} \times \text{Selling Price}$$

3) Data Validation

After populating each table, select queries were used to verify that the records had been inserted correctly. Foreign key relationships were maintained throughout the data population process.

For example:

```
select * from production;
```

was used to verify the production records, while:

```
select * from sale;
```

was used to verify the sales transactions.

The populated database is now ready for the development and testing of SQL queries addressing practical agribusiness business questions.

SQL Query Development

Query 1: Total Production by Crop

Business question:

Which crops have the highest total production?

SQL query:

```
select
  c.crop_name,
  sum(p.production_quantity) as total_production
from production p
join crop c
  on p.crop_id = c.crop_id
group by c.crop_name
order by total_production desc;
```

Explanation:

This query calculates the total production for each crop. The production table is joined with the crop table to display the crop name. The sum() function adds the production quantity for each crop. The results are grouped by crop and arranged from highest to lowest production.

Result:

Wheat has the highest total production at 1,175.00, followed by Rice at 905.00. Mustard has the lowest total production at 97.50.

Screenshot:

	crop_name	total_production
▶	Wheat	1175.00
	Tomato	495.00
	Sugarcane	640.00
	Soybean	232.00
	Rice	905.00
	Potato	275.00
	Onion	240.00
	Mustard	97.50
	Maize	520.00
	Cotton	230.00

Query 2: Total Production by Farm

Business question:

Which farms have the highest total production?

SQL query:

```
select
    f.farm_name,
    sum(p.production_quantity) as total_production
from production p
join farm f
    on p.farm_id = f.farm_id
group by f.farm_name
order by total_production desc;
```

Explanation:

This query calculates the total production for each farm. The production table is joined with the farm table to display the farm name. The sum() function adds the production quantities for each farm. The results are grouped by farm and arranged from highest to lowest production.

Result:

Sunrise Agro Farm has the highest total production at 1,340.00. Deccan Agro Farm has the lowest total production at 132.00.

Screenshot:

	farm_name ▼	total_production
►	Sunrise Agro Farm	1340.00
	Nashik Harvest Farm	465.00
	Haryana Fields Farm	562.50
	Green Valley Farm	635.00
	Golden Harvest Farm	555.00
	Ganga Plains Farm	745.00
	Deccan Agro Farm	132.00
	Agra Green Farm	375.00

Query 3: Total Production by Year

Business question:

How much total production was recorded in each year?

SQL query:

```
select  
  
    production_year,  
  
    sum(production_quantity) as total_production  
  
from production  
  
group by production_year  
  
order by production_year;
```

Explanation:

This query calculates the total production for each year. The `sum()` function adds the production quantities, while `group by` groups the records according to the production year. The results are displayed in chronological order.

Result:

Total production was 1,942.50 in 2024 and increased to 2,867.00 in 2025.

Screenshot:

	production_year	total_production
▶	2025	2867.00
	2024	1942.50

Query 4: Total Cost by Production Record

Business question:

What is the total cost associated with each production record?

SQL query:

```
select
    production_id,
    sum(amount) as total_cost
from cost
group by production_id
order by total_cost desc;
```

Explanation:

This query calculates the total cost for each production record. The `sum()` function adds all cost amounts belonging to the same production record. The results are grouped by `production_id` and arranged from highest to lowest total cost.

Result:

Production record 7 has the highest total cost of 16,500.00, while production record 2 has the lowest total cost of 4,500.00.

Screenshot:

	production_id	total_cost
▶	7	16500.00
	13	14500.00
	1	14000.00
	9	12500.00
	18	12000.00
	4	10800.00
	10	9000.00
	3	8500.00
	20	7500.00
	5	7000.00
	8	7000.00
	14	6500.00
	19	6500.00
	16	6200.00
	15	5300.00
	12	5200.00
	17	5100.00
	6	5000.00
	11	4800.00

Query 5: Total Cost by Input

Business question:

Which inputs have the highest total usage cost?

SQL query:

```
select
    i.input_name,
    sum(iu.cost) as total_cost
from input_usage iu
join input i
    on iu.input_id = i.input_id
group by i.input_name
order by total_cost desc;
```

Explanation:

This query calculates the total cost associated with each input. The input_usage table is joined with the input table to display the input name. The sum() function adds the costs for each input. The results are grouped by input and arranged from highest to lowest cost.

Result:

DAP Fertilizer has the highest total usage cost at ₹39,000.00, while Herbicide has the lowest at ₹3,900.00.

Screenshot:

	input_name ▲	total_cost
▶	DAP Fertilizer	39000.00
	Herbicide	3900.00
	NPK Fertilizer	19250.00
	Organic Compost	9300.00
	Pesticide	13050.00
	Rice Seed	11750.00
	Urea Fertilizer	38400.00
	Wheat Seed	15250.00

Query 6: Revenue by Market

Business question:

Which markets generate the highest total sales revenue?

SQL query:

```
select
    m.market_name,
    sum(s.revenue) as total_revenue
from sale s
join market m
    on s.market_id = m.market_id
group by m.market_name
order by total_revenue desc;
```

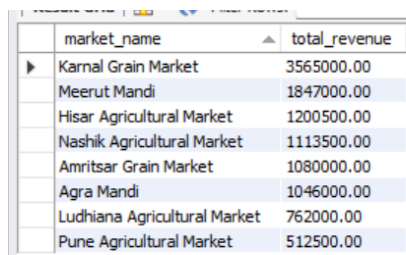
Explanation:

This query calculates the total sales revenue generated through each market. The sale table is joined with the market table to display the market name. The sum() function adds the revenue from all sales for each market. The results are grouped by market and arranged from highest to lowest revenue.

Result:

Karnal Grain Market generated the highest total revenue at ₹35,65,000.00, while Pune Agricultural Market generated the lowest at ₹5,12,500.00.

Screenshot:



market_name	total_revenue
Karnal Grain Market	3565000.00
Meerut Mandi	1847000.00
Hisar Agricultural Market	1200500.00
Nashik Agricultural Market	1113500.00
Amritsar Grain Market	1080000.00
Agra Mandi	1046000.00
Ludhiana Agricultural Market	762000.00
Pune Agricultural Market	512500.00

SQL Testing and Optimization

Query Testing

The SQL queries were tested in MySQL Workbench after data population. A record count check was also performed to confirm that the production table contained the expected 20 records.

```
select count(*) as production_records  
from production;
```

The query returned 20 records, confirming that the production data was populated correctly.

Query Execution Analysis

The explain command was used to examine how MySQL executes the crop production query.

```
explain  
select  
    c.crop_name,  
    sum(p.production_quantity) as total_production  
from production p  
join crop c  
    on p.crop_id = c.crop_id  
group by c.crop_name  
order by total_production desc;
```

The execution plan showed that MySQL uses the crop_id index when accessing the production table. The crop table contains only 10 records, so scanning this small table is reasonable. MySQL also uses a temporary result and sorting for the grouping and ordering operations.

Since the database currently contains a small amount of sample data, the query performs efficiently without requiring additional indexes. The existing primary keys and foreign-key-related indexes are sufficient for the current workload.

Conclusion:

The queries were tested successfully, and explain was used to verify the execution plan. No additional optimization was required for the current dataset.

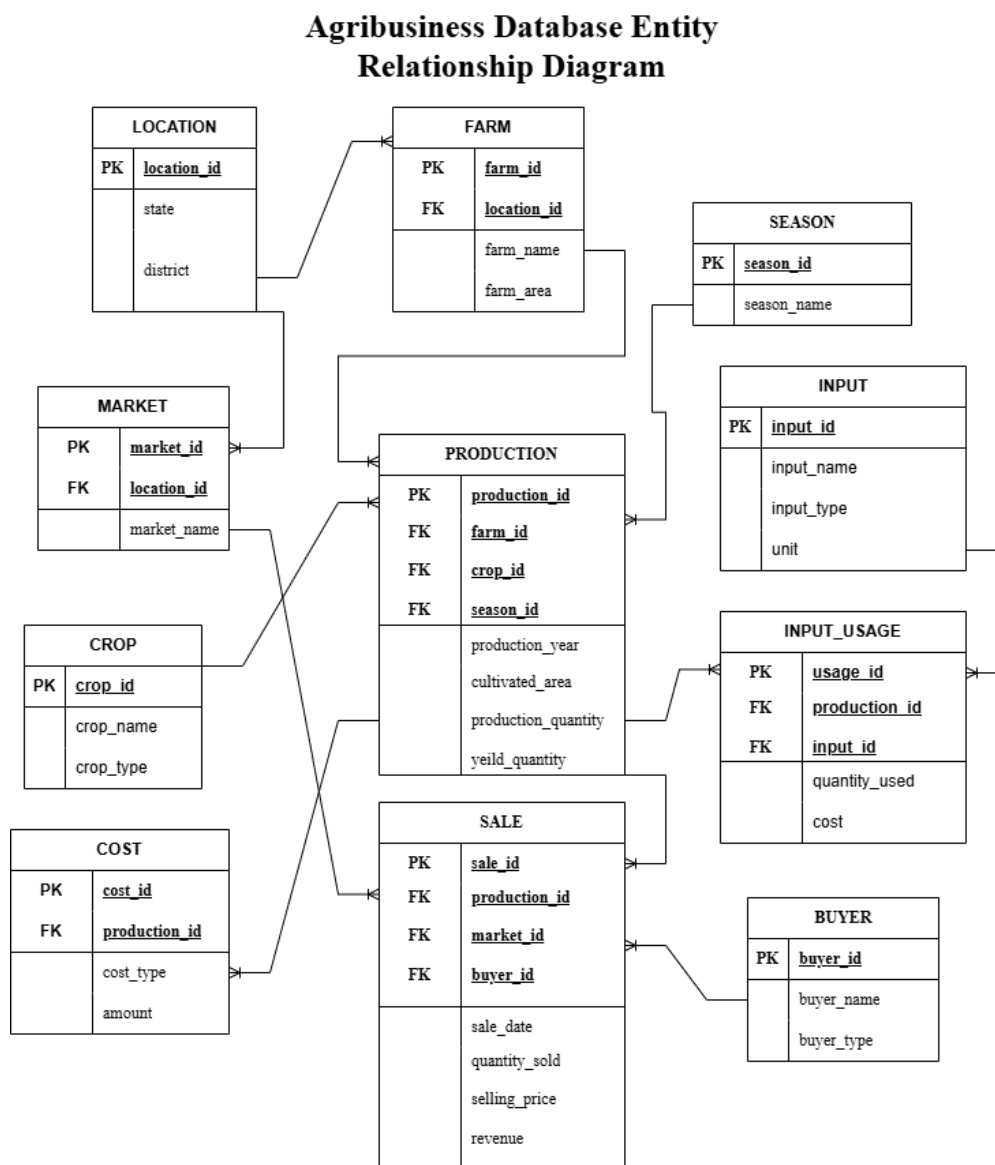
Entity Relationship Diagram

The Entity Relationship Diagram (ER diagram) represents the structure of the agribusiness database and shows how the different tables are connected. The model was designed to support farm management, crop production, input usage, cost tracking, sales, markets, buyers, seasons, and business performance analysis.

The diagram contains 11 main entities: Location, Farm, Crop, Season, Production, Input, Input Usage, Cost, Market, Buyer, and Sale. Primary keys are used to uniquely identify records, while foreign keys establish relationships between related tables.

ER Diagram

The following diagram shows the complete relational structure of the agribusiness database.



Relationship Structure

The database uses one-to-many relationships between the main entities:

- One location can have many farms.
- One farm can have many production records.
- One crop can appear in many production records.
- One season can be associated with many production records.
- One production record can have many input usage records.
- One input can be used in many production records through the Input Usage table.
- One production record can have multiple cost records.
- One location can have many markets.
- One production record can have multiple sale records.
- One market can have many sale transactions.
- One buyer can be associated with many sale transactions.

The Production table acts as a central table for crop and farm-related activities. The Input Usage and Sale tables connect production activities with inputs and market transactions, while the Cost table records different expenses associated with production.

Database Support for Common Business Inquiries

The database was designed to support common agribusiness questions related to crop production, farm performance, costs, inputs, and market sales. The relationships between the tables allow information from different parts of the business to be combined and analyzed using SQL.

Crop Production Analysis

The production table stores production quantity, cultivated area, yield, production year, farm, crop, and season information. By joining production with the crop table, the total production of each crop can be calculated and compared.

This supports questions such as:

- Which crops have the highest total production?
- How does production vary between different crops?

The analysis showed that Wheat had the highest total production in the sample dataset.

Farm Performance Analysis

The farm and production tables are connected through farm_id. This allows production records to be grouped by farm and used to compare farm-level production.

This supports questions such as:

- Which farms have the highest total production?
- How much production is associated with each farm?

The analysis showed that Sunrise Agro Farm had the highest total production among the farms in the dataset.

Time-Based Production Analysis

The production table contains production_year, allowing production to be grouped and compared across years.

This supports questions such as:

- How does total production change over time?
- Which year recorded higher production?

In the sample data, total production increased from 1,942.50 in 2024 to 2,867.00 in 2025.

Cost and Input Analysis

The cost table stores individual production costs, while input_usage records the inputs used, their quantities, and associated costs. These tables can be linked with production and input to analyze production expenses and input usage.

This supports questions such as:

- What is the total cost associated with each production record?
- Which inputs have the highest usage cost?
- Which production activities require higher expenditure?

The analysis showed that Production ID 7 had the highest recorded total cost, while DAP Fertilizer had the highest total usage cost among the inputs.

Market and Sales Analysis

The sale table records quantity sold, selling price, revenue, sale date, production, market, and buyer information. The market table allows sales to be grouped by market.

This supports questions such as:

- Which markets generate the highest sales revenue?
- How much revenue is generated through each market?
- Which buyers are associated with sales transactions?

The analysis showed that Karnal Grain Market generated the highest total sales revenue in the sample data.

Overall Business Analysis

By combining the related tables, the database provides a structured way to analyze production, farm operations, inputs, costs, and sales together. This allows the database to support both operational questions, such as production and input usage, and business questions, such as costs and market revenue.

The relational structure also avoids storing the same information repeatedly across different tables. For example, crop details are maintained in the crop table, while production records refer to them through crop_id. This makes the database organized and suitable for SQL-based analysis.

Profitability can also be analysed by comparing sales revenue with the associated production and input costs.

Conclusion

The Week 2 project focused on designing a relational database for agribusiness operations and developing SQL queries for business analysis. The database was structured into related tables covering locations, farms, crops, seasons, production, inputs, costs, markets, buyers, and sales.

The data was populated successfully and the relationships between the tables were tested through SQL queries. Six business-focused queries were developed to analyze crop production, farm performance, yearly production, production costs, input costs, and market revenue.

The queries were tested in MySQL Workbench, and the explain command was used to examine query execution. The execution plan was considered appropriate for the current dataset, and no additional optimization was required.

Overall, the database provides a structured foundation for analyzing agribusiness operations and answering common business questions using SQL.